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Investigating Students' Emotions in Virtual Reality with GenAI based RAG Implementation

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Abstract: This case study examined the emotional responses of students engaging with extended reality and generative AI (GAI) technologies in a learning environment. Twelve fourth-grade students participated in an eight-day STEAM program supported by advanced technologies. During the program, the students were tasked with creating and organizing a virtual ecology corner in their classroom utilizing virtual reality (VR), which encompassed both immersive virtual reality (IVR) and mixed reality (MR). The study assessed students' emotional experiences while interacting with various technologies during instruction. Findings revealed that students generally felt positive emotions, such as enjoyment and hope, regardless of the specific technology employed. However, higher functional complexity of the technology correlated with increased levels of student anxiety. Additionally, girls exhibited slightly more anxiety than boys throughout the learning activities.

Introduction

There is a growing interest in exploring emotions in learning, which has been historically marginalized compared to cognitive aspects (Dukes et al., 2021; Hayat et al., 2020; Richards, 2022). Students' emotions can impact their learning through attention, memory, cognition, and meta-cognition processes (Obergriesser & Stoeger, 2020; Taub et al., 2021), which can further impact students' learning outcomes (Chang et al., 2023; Hayat et al., 2020). Therefore, understanding emotions is crucial for gaining insights into human learning.

As the learning environment can significantly influence the emotions of learners (Li et al., 2020; Taub et al., 2021), it is important to investigate the potential impact of technology-supported learning environments on students' emotions. Recently, there has been increasing attention on the potential of innovative technologies such as extended reality (XR) and generative AI (GAI) that could revolutionize classrooms (Baidoo-Anu & Ansah, 2023; Logeswaran et al., 2021). Despite the beneficial features of XR and GAI in enhancing learning, few studies have examined students' emotions in learning environments that incorporate both technologies.

Therefore, this study examined students' emotions while engaging in a learning environment that incorporated extended reality and generative AI (XRGAILE). This study has two research questions:

1. What emotions did students experience when using different technologies, i.e., immersive virtual reality (IVR), mixed reality (MR), XR (in this case, a combination of IVR and MR) and GAI?
2. What factors influence students' emotions within an XRGAILE?

Research background

Academic emotions and learning

Pekrun's Control-Value Theory (CVT) is a key theory in educational research (Pekrun, 2006). Based on the CVT theory, academic emotions are tied directly to learning activities or learning outcomes (Pekrun, 2006). These emotions can be categorized as positive and negative, activating and deactivating, and activity-oriented and outcome-oriented (Lewis et al., 2010). From the CVT perspective, emotions are linked to motivations based on how learners perceive their control and value in their practice (Pekrun, 2017).

Academic emotions can impact students' learning outcomes through attention, motivation, use of learning strategies, and self-regulation of learning (Pekrun, 2017). Many studies have shown that positive emotions such as enjoyment can positively predict students' academic performance (e.g., Putwain et al., 2021), cognitive strategies use (Obergriesser & Stoeger, 2020), and motivation during the learning process (Pekrun, 2017). However, negative emotions have mixed effects on students' learning. For example, while boredom and anxiety can negatively predict subsequent achievement (Putwain et al., 2021), frustration can be a beneficial emotion for predicting individuals' cognitive performance in positive ways (Taub et al., 2021).

Emotions in technology-supported learning environments

Regarding the development of emotions, it was found that factors such as learning environment, prior achievement, gender, and socioeconomic status can impact students' emotions (Pekrun, 2017). Hence, technology-

supported learning environments can influence students' emotions. In this study, we incorporated XR and GAI into the learning environment. XR includes virtual reality (VR), mixed reality (MR), and augmented reality (AR). Specifically, as a type of VR, IVR can provide students with a 360-degree immersive experience through IVR devices; MR encompasses the integration of virtual and physical environments, allowing for interactive experiences within both realms; and AR superimposes digital data and objects onto the real world (Logeswaran et al., 2021; Maas & Hughes, 2020). Previous studies indicate that while learning within IVR (Liu et al., 2022), AR (López-Faican & Jaen, 2020), and MR (Beyoglu et al., 2020), primary school students experienced higher levels of positive emotions (e.g., enjoyment and curiosity) and lower levels of negative emotions (e.g., anxiety), suggesting XR's benefits for learning.

Another technology used in this study, GAI, is an innovative AI technology based on large language models. It can generate continuous, multimodal, personalized responses and feedback based on students' input, potentially supporting learning activities (Baidoo-Anu & Ansah, 2023). This capability makes GAI a promising tool for enhancing educational experiences. Up till now, few studies have explored students' emotions when learning with GAI. Understanding these emotional experiences is crucial for optimizing the use of GAI in educational settings.

Methods

Learning environment: XRGAILE

The XRGAILE involved IVR, MR, and GAI. For IVR technologies, we used an XR application called *MultiBrush*. This application allows students to view 360-degree panoramic pictures (see Figure 1a) and draw 3D objects collaboratively (see Figure 1b) via the IVR devices. Students can scale and rotate the object, which allows them to draw paintings from different scales, angles, and perspectives. This can afford students an expanded opportunity for imagination and creativity. Regarding MR technologies, we used an XR application developed based on an open-source VR application called *OpenBrush*. Students can use OpenBrush to interact with their VR paintings within a real-world setting (see Figure 1c for an example).

Figure 1

AI-generated panoramic pictures (a), students' 3D painting in Multibrush (b) and Openbrush (c)

(a) (b) (c)

We used two GAI applications called *SkyBox AI* to generate panoramic pictures and *Midjourney* to generate general pictures according to the prompts. Figure 1a showed an AI-generated panoramic picture with the prompt "realism, ecological wetland, flowers, birds, fish, insects, forest, iceberg." Students also used a GAI application called *ChatGPT* to generate texts during their creative activities. When using the GAI application, students wrote the prompts, and the teacher or research assistants then helped them send these prompts to the GAI.

Learning activities and procedures

We developed a series of STEAM activities based on the design thinking approach (Henriksen et al., 2017). The STEAM activities are aimed at developing students' creativity, understanding of ecological concepts, and appreciation of art. The main task for students is to design an ecological corner for their classroom, which is a complex, ill-structured problem. Here, the "ecological corner" refers to a designated space within a regular classroom where students design and set up a combination of small animals, plants, and microorganisms. Each setup is required to be based on a theme chosen by the students, such as a succulent garden or a miniature Jurassic Park, and can range from a few potted plants to more elaborate arrangements. Students need to define, ideate, prototype, and assess their designs, and finally develop their creativity through hands-on practices (Amabile & Pratt, 2016; Huang et al., 2024). The main focus of the STEAM activities is art, but they also incorporate science, as students are encouraged to consider how to keep the organisms alive by applying their scientific knowledge and thinking. Technology is also integrated, as students created and refined their design within a XRGAILE.

The learning activities began with a pretest followed by an eight-day instructional program. Students were divided into two groups based on gender. Each group was then split into sub-groups of three students. Each session was 60 minutes long, during which the students collaborated on daily tasks. The program was facilitated by an art teacher. After completing all eight sessions, the post-test was administered to all students. Following the post-test, the students participated in focus groups.

On Day 1, students followed the teacher's instructions to explore VR painting using IVR devices. On Day 2, students received their main task: to design an ecological corner for the classroom. To spark their creativity, students used Skybox AI to generate panoramic pictures and viewed these pictures in the IVR-based environment.

On Day 3, students began creating the draft of their design in MultiBrush. Students were asked to use Midjourney to generate general pictures to support their design. Students viewed these AI-generated pictures through a tablet. After finishing the draft, the teacher helped students set up their draft prototype in the designated space in the classroom through OpenBrush. On Day 4, the teacher provided instructions and feedback on the drafts and taught students relevant knowledge and skills. After that, all groups revised their drafts solely using paper and pencils. On Days 5 and 6, students continued to refine and finalize their designs under the teachers' guidance.

On Day 7, students wrote a brief introduction to describe their work. To stimulate their creative thinking, students used ChatGPT to generate a draft description for their work. The teacher again assisted in setting up their VR paintings in the classroom through OpenBrush. Finally, on Day 8, each group presented their design to the whole class. The teacher and all the students discussed and evaluated each design. Students completed their daily worksheets and questionnaires during the learning activities.

Data

This case study involved 12 Year 4 students (six boys and six girls) from a public primary school in Shanghai. Most participants had no prior experience with XR or GAI, except for two students who reported owning VR devices at home. The data was collected during the fall semester of 2023. We collected students' (a) demographic information, (b) pre-and post-tests to evaluate their creativity, (c) process data that recorded academic emotions during learning, and (d) a focus group interview after completing all sessions. Students' emotions were assessed at the end of the daily session except on Day 4 and Day 8, when no technologies were used. We evaluated students' emotions while using IVR, MR, XR, and GAI based on the Achievement Emotions Questionnaire (AEQ) with a five-point Likert scale (Pekrun et al., 2011). The mean Cronbach's α of the AEQ is 0.8.

Results

RQ1: What emotions did students experience when using different technologies?

As shown in Table 1, students experienced high levels of positive emotions ($M_{positive}$ = 4.43, $SD_{positive}$ = 0.90) and low levels of negative emotions ($M_{negative}$ = 1.16, $SD_{negasitive}$ = 0.53) throughout the learning activities. To further explore the emotions students experienced in the XRGAILE and the factors that triggered them, we created a Bayesian generalized linear model (GLM) with emotional intensity as the dependent variable. The Bayesian GLM was chosen for its flexibility in modeling data that deviate from normality and its suitability for small sample sizes. Independent variables include the type of emotions, the type of technology, gender, and some of their interactions, while learning duration (as the ordinal variable "Day") was treated as a random effect to account for variability across learning days. Anger and IVR were set as the baseline levels.

Table 1

Students Emotions when Learning with Different Technologies

Category	Emotion Type	IVR	MR	XR	GAI
Positive Emotions	Enjoyment	4.48 (0.63)	4.50 (1.17)	4.50 (0.80)	4.53 (0.63)
	Hope	4.51 (0.62)	4.67 (0.65)	4.50 (1.24)	4.54 (0.64)
	Pride	4.23 (0.99)	4.33 (1.37)	4.25 (1.14)	4.19 (1.05)
Negative Emotions	Anger	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.06 (0.19)
	Anxiety	1.56 (0.61)	1.00 (0.00)	1.25 (0.45)	1.21 (0.28)
	Shame	1.27 (0.38)	1.00 (0.00)	1.08 (0.29)	1.14 (0.22)
	Hopelessness	1.10 (0.29)	1.25 (0.87)	1.00 (0.00)	1.06 (0.19)
	Boredom	1.12 (0.25)	1.00 (0.00)	1.08 (0.29)	1.08 (0.29)

RQ2: Factors that influence students' emotions within the XRGAILE

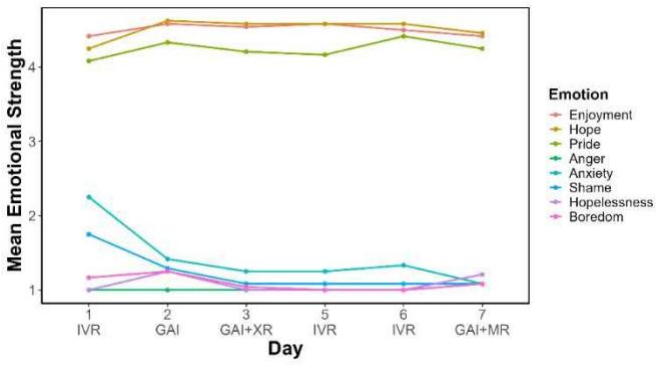
The results indicated that compared to the baseline emotional intensity of anger, students experienced significantly higher emotional intensity when feeling enjoyment (β = 10.19, 95% CI [7.23, 13.97]), hope (β = 10.38, 95% CI [7.42, 14.19]), pride (β = 9.52, 95% CI [6.61, 13.40]), anxiety (β = 6.21, 95% CI [3.25, 10.00]), and shame (β = 5.18, 95% CI [2.22, 9.00]), regardless of the type of technology they used. The result of the focus group is aligned with the quantitative results. In the focus group, the students reported their emotions as "interested," "excited," "happy," "surprised," and "new." They gave an example that they were surprised by using a VR brush to draw flower petals.

Regarding the learning duration, the random effect (SD = 0.29, 95% CI [0.02, 0.78]) suggests moderate variability in emotional intensity across different days. As shown in Figure 2, there is a decreasing trend in students' anxiety and shame over the course of daily learning activities. Additionally, students' pride fluctuated over the days, with increases on Day 2 and Day 6. In the focus group, a girl said she was anxious when her drawing did not meet her expectations and had to repeat several times. However, she was pleased with the final product after several attempts.

Regarding the type of technology, the results indicated that students experienced marginally lower emotional intensity when feeling anxious (β = -2.61, 95% CI [-6.28, 0.95]) while using GAI compared to IVR. Furthermore, girls experienced marginally higher levels of anxiety (β = -2.59, 95% CI [-6.09, 0.64]) and shame (β = -2.54, 95% CI [-6.11, 0.87]) than boys. In the focus group, three students mentioned feeling anxious while doing VR painting due to the complexity of the functions. One girl recalled, "Because it is easy to make mistakes when drawing, we always have to use an eraser. We cannot withdraw it in a multi-person room, and it will always erase other drawings. I had to draw over and over again."

Figure 2

Students' Emotions over the Daily Learning Activities



Scientific and educational importance of the study

The preliminary findings of our study revealed several important insights and suggestions. First, students experienced positive emotions throughout the learning activities, with a low degree of negative emotions, suggesting that the XRGAILE provides a positive learning experience. Second, half of the students reported feeling more anxious when using IVR, likely due to the challenges of mastering the complex functions of VR painting. However, anxiety and shame decreased over time, suggesting the importance of designing XR and GAI learning activities with sufficient duration to allow students to adapt to novel technologies. Fourth, students' emotions, such as pride, fluctuated during the learning activities, even when using similar technologies, suggesting that the daily tasks might impact their emotions, warranting further exploration in future analysis. Finally, girls appeared slightly more anxious than boys, indicating a need for additional emotional support for female students when engaging with innovative technologies like XR and GAI. Nevertheless, as a preliminary study with only 12 participants, generalizing these findings to a larger population needs to be cautious.

Overall, this study reveals that an XR and GAI-supported learning environment has the potential to facilitate positive learning experiences. The design of learning environments and activities of this study offers a model for supporting students' positive emotions in learning with innovative technologies.

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